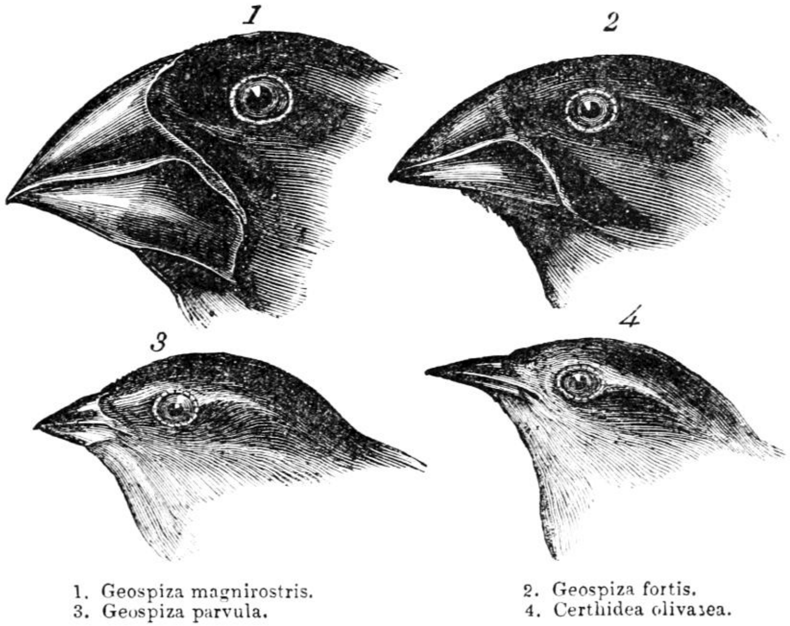
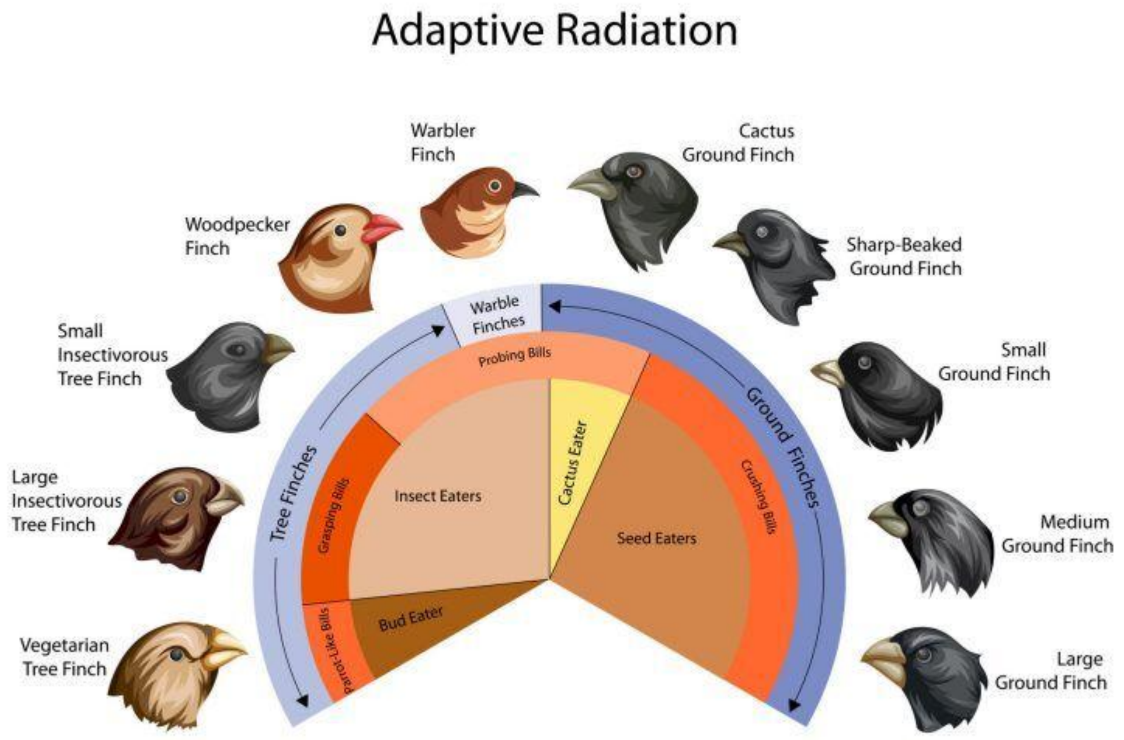
A simulation model for trait-mediated interspecific competition in birds



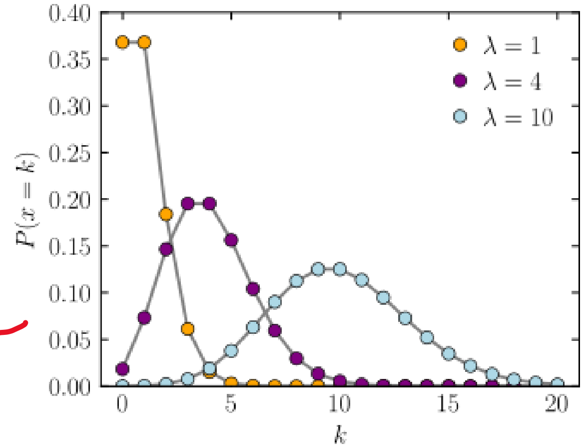
Jimmy Choi, McTavish Lab

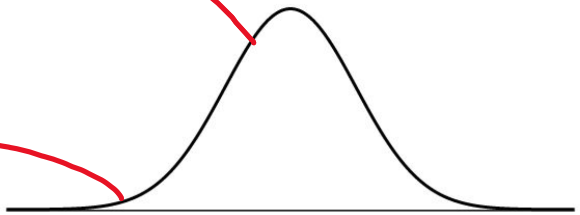
We may love birds but they don’t love each other

- *Aves*: well-studied, charismatic taxa
- Even Darwin cut his teeth on them
- Considering evolutionary principles and timescales alongside ecological ones (eco-evo model)



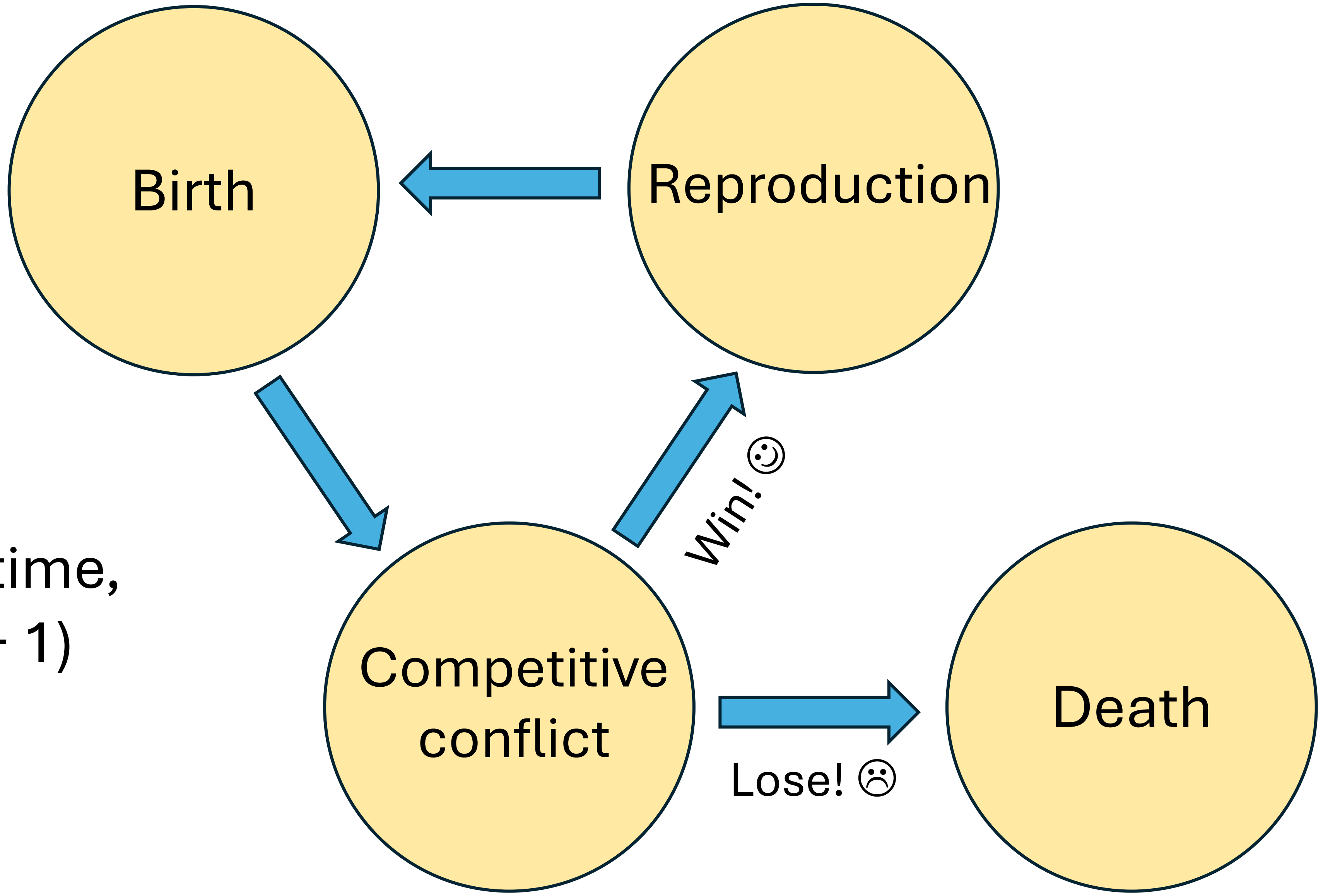
So how do you model evolution?





Probability!

Model overview (assumed flow of events)

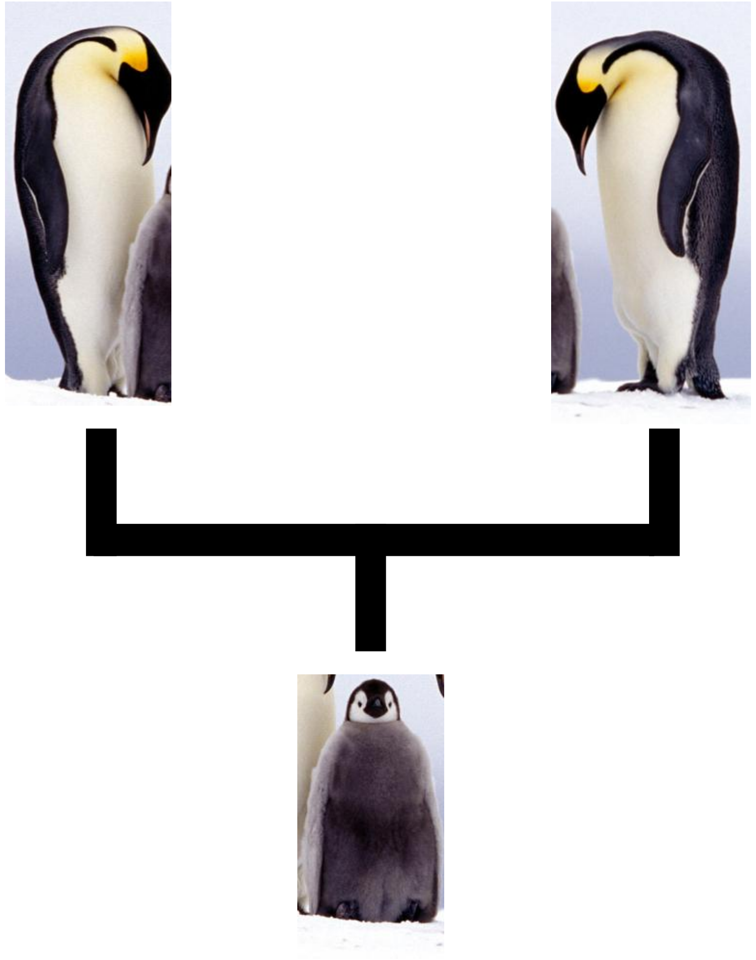


- Discrete time, $n(t) \rightarrow n(t + 1)$

Finer genetic details

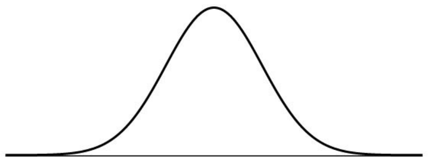
trait value
$z_i = 1.2$

trait value
$z_i = 1.0$



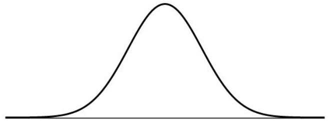
trait value
$z_{offspring} = [(1.2 + 1.0)/2] + \epsilon$

- After the “competitive conflict” phase, each survivor from both species can breed & reproduce
- Trait value of offspring is determined by “mating” system, two random individuals within each species are matched
- Average of both parents’ trait values are taken, then ϵ (mutation/developmental noise term) is added
- ϵ is normally distributed with a mean at 0 and a standard deviation which we call “mutation rate” for simplicity



Finer ecological details

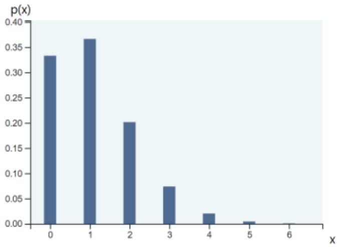
- Traits are normally distributed around a provided mean and stdev



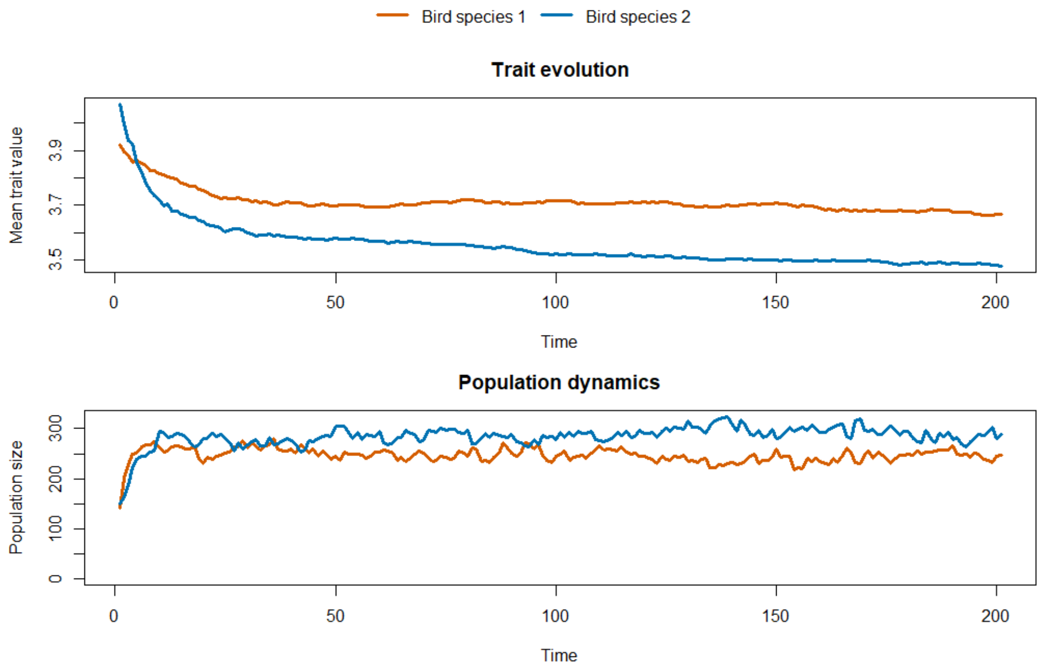
- An individual bird's trait value is compared against the environment's selective trait optimum & selection surface, "weighted coin flip"

$$P(win \mid z_i, z^{*}) = \Phi[(z_i - z^{*}) / \sigma_s]$$

- Loss means death, winning means being able to breed & bear offspring (via Poisson distribution w/ λ = 1.1, density-dependent)

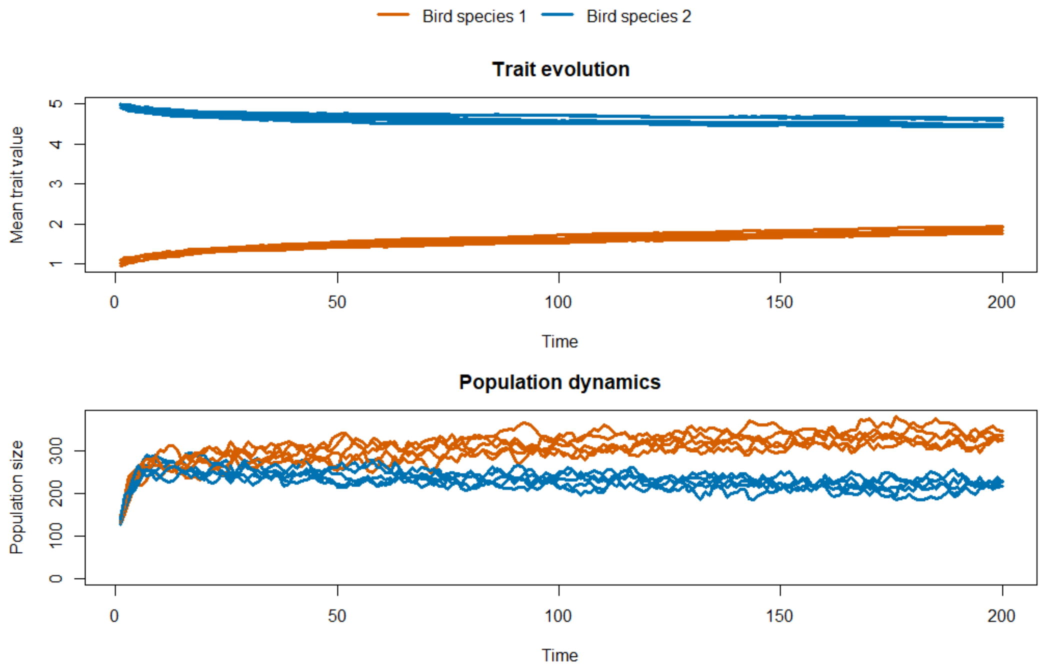


A useful model for visualizing eco-evo concepts

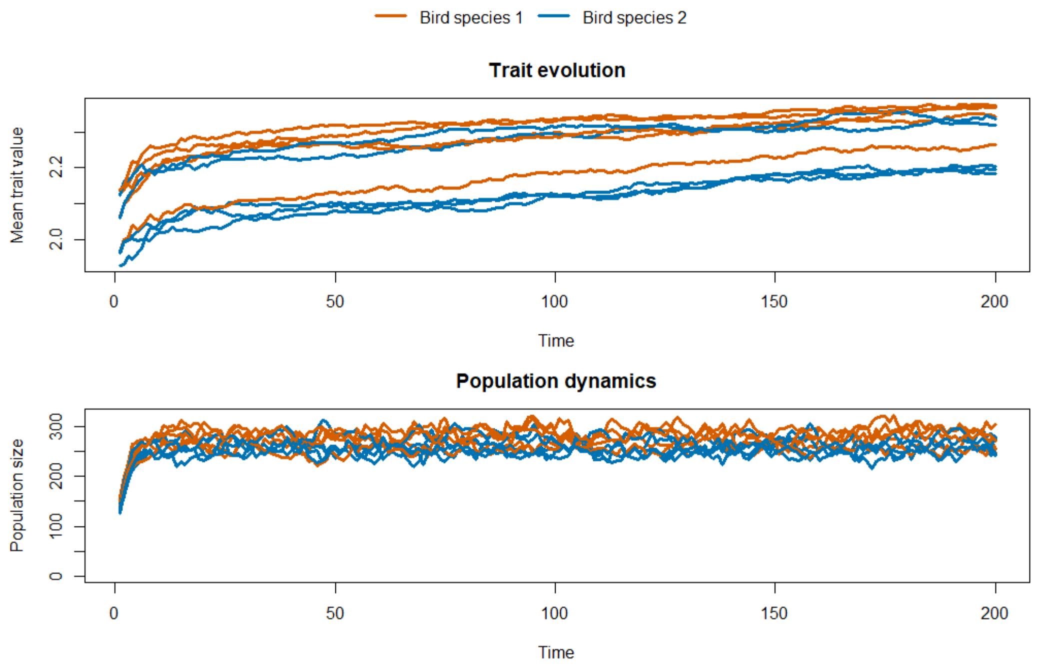


Scenario 1. Both species start at 100 individuals with a trait value mean of 4. The standard deviation of the trait in species 1 is 0.5, while in species 2 it is 0.8. Single replicate. *(Stabilizing selection & how genetic variation benefits adaptation)*

Scenario 2. Both species start at 100 individuals and are equally far away from the trait optimum (3.0) in opposite directions, with equal trait standard deviations. Species 1 has a "mutation rate" (stdev of ϵ) of 0.1, while in species 2 it is 0.05. 5 replicates were conducted. *(Mutation as "fuel" for evolution)*

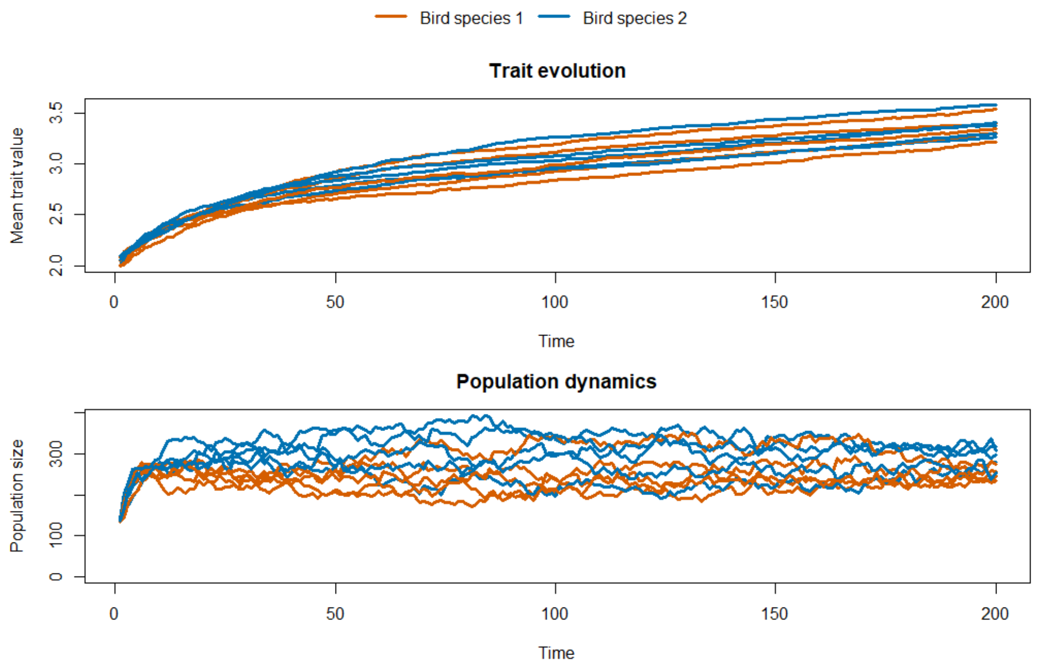


A useful model for visualizing eco-evo concepts



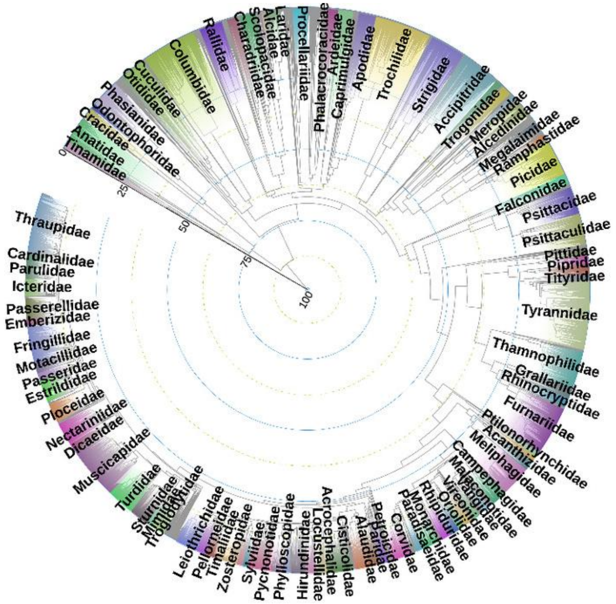
Scenario 3. Both species start at 100 individuals with a trait value mean of 2, equal genetic variation, and equal "mutation rates", with an environmental optimum of 5. The width of the selection surface (the inverse of selection strength) was set to 2.0. 5 replicates were conducted. *(Weak selection)*

Scenario 4. Identical to Scenario 3, except the width of the selection surface was set to 0.5. *(Strong selection)*



Thanks for listening!

Try the model yourself
by scanning the QR code!



Family-level bird phylogeny adapted from McTavish et al. (2025).

A great many thanks to Justin for letting me peek at his code and teaching materials.